CMSC 5718 – Practice Questions

Topic 0-1

1. [slide 42, 43] An initial investment of $1000 becomes $1230 after 2.5 years. Express the annualized return of this investment in (a) the annual compounding convention; (b) the continuous compounding convention.

Topic 1-1

2. [slide 10] Spot gold is trading at US$4780. The 6-month USD interest rate is 4.18% p.a.. Find the theoretical price of a gold futures contract with a maturity of 6 months (=0.5 year).

3. [slides 13-14] Assume the following market rates:
   Spot FX: EUR 1 = USD 1.0782, 6-month forward FX rate = 1.0927 (market price),
   USD 6-month interest rate is 5.13% p.a., EUR 6-month interest rate is 3.75% p.a.
   Identify an arbitrage trade from these market rates, and calculate the theoretical arbitrage profit.

4. [slides 36-37] Assume the following option prices and market parameters:
   Stock price = $50, strike = $48, 3-month interest rate = 4.78%p.a., maturity = 0.25
   Market price of call option = $5.33, market price of put option = $2.26
   Identify an arbitrage trade from these given prices, and calculate the theoretical arbitrage profit.

Topic 1-2

5. [slides 5-8] Assume the current stock price is $88. According to a forecast, after 6 months there is a 75% probability that the stock price will become $100, and there is a 25% probability that the stock price will become $70. The 6-month risk free rate is 4.82%p.a.
   a. What is the expected price of the stock under pricing by expectation?
   b. What is the expected price of the stock under pricing by no arbitrage?

Topic 1-3

6. [slides 29, 31] An investor bought a bull ELN and paid 97.5% of the nominal amount. The initial maturity is 3 months. Now with a remaining maturity of one month (=1/12 year), the stock price has gone up and the investor is thinking about redeeming the ELN early. The investment bank uses the following parameters:
   1-month interest rate = 4.38%p.a., option price = 0.175% of nominal amount,
   commission charged by the investment bank for early redemption = 0.30%.
   What is the new note price that the investment bank will pay to buy back the ELN from the investor?

7. [slide 37] A Capital Guaranteed Note has the following characteristics:
   Guaranteed level G: 100% of N, initial amount X=$100, tenor T=3 years, Interest rate r = 5.3%p.a., option premium P=18.7% (from option pricing model),
   Participation level = 0.62, initial selling price = 100% of notional amount N
   Calculate the initial commission charged by the bank.

8. [c.f. slide 42] A structured product is constructed with the following components:
   Long a bond that pays 100% at maturity, short a put spread, long a digital option that pays 2% if the spot price is above 107% of the initial stock price.
   If the investment bank wants to keep the same selling price for this product, but wants to increase its profit margin, suggest what possible changes need to be made for the characteristics of the digital option.

Topic 1-4

9. [slide 33] You have a long position in a 1-year European call, strike 100, with the following risk parameters: delta = 0.6115, gamma = 0.0128, vega = 0.3832. Two other options are available:
   - 6-month European put, strike 95, delta = -0.3364, gamma = 0.0156, vega = 0.2580
   - 18-month European call, strike 105, delta = 0.5809, gamma = 0.0114, vega = 0.4785

   In order to keep the portfolio delta, gamma, and vega neutral at the same time, assume a is the required position in the strike 95 put, b is the required position in the strike 105 call, and c is the required position in the underlying shares. Find a, b, and c.

10. [slide 36] The current stock price is $100. A portfolio with options on this stock has the following Greeks: delta shares = -4300, vega = $-4000 (for +1 % move in volatility).
    a. If the share price has moved to $98, and the volatility level has shifted up by 2%, estimate the change in the profit/loss of the portfolio.
    b. If the share price has moved to $88 and the volatility level remains unchanged, can you still estimate the change in the profit/loss with the method used above?

Topic 2-1

11. [slides 24, 26, 29] Given the following s.d.'s for the returns of stocks 1 and 2: $\sigma_1 = 25\%, \sigma_2 = 32\%$.
    a. Find the portfolio s.d. if $w_1 = 0.4, w_2 = 0.6, \rho_{12} = 0.3$.
    b. Find the portfolio s.d. if $w_1 = 0.4, w_2 = 0.6, \rho_{12} = -1$.
    c. If the correlation coefficient $\rho_{12} = 0$, find the weights $w_1$and w_2 when the portfolio s.d. is at a minimum.

12. [slide 47] A is a risky portfolio and B is a risk free asset, and the following parameters are given: $r_A = 12\%, \sigma_A = 33\%, r_B = 4.5\%$. If a portfolio is formed from A and B and the portfolio s.d. $\sigma_P = 21\%$, find the return of this portfolio.

13. [slides 62-63] An investor has a utility function given by: $u(x) = \sqrt{x}$ (where *x* is the worth of the investment).
    a. The expected utility of an investment scheme A is $260. Find the certainty equivalent of this investment.
    b. Another investment scheme B is available, where there is a 60% chance of returning $100000 and 40% chance of returning $20000. Should the investor choose A or B, if the investor wants to maximize the utility achieved?

14. [c.f. slide 66] An investor adopts a quadratic utility function with the form: $u(x) = 36x - 2x^2$, where x is the wealth level. Derive the applicable range of x values of which this function is applicable.

Topic 2-3

15. [slide 8] The return of a risk free asset is given as: $r_f = 5\%$.
    a. If a stock has a beta $\beta_1 = 0.38$ and its equilibrium return $\overline{r_1} = 7\%$, find the return of the market portfolio r_M.
    b. Another stock has a beta $\beta_2 = 1.32$. Find its equilibrium return $\overline{r_2}$ using the r_M obtained above.

16. [slides 12, 13, 16] The risk free rate is 5.5%, and the following parameters are given:

Portfolio	Rate of return (%)	Standard deviation	Beta
A	11%	10%	0.5
Market Portfolio	14%	12%	1.0

    a. Does portfolio A lie above or below the CML?
    b. Does portfolio A lie above or below the SML?

17. [slide 17] A firm is considering a project with a beta of 1.55. This project has an expected return of 18%. The risk free rate is 4.8% and the expected return of the market portfolio is 13%. Should the project be accepted?

18. [slide 30] The 1-year target price of the stock $\bar{Q}$ is \$130 and the beta β = 1.41. Expected return of the market portfolio $\overline{r_M}$ = 13% and the risk free rate r_f = 4.3%. If the stock is trading at \$114.8 today, is this a good investment opportunity according to CAPM?

Topic 3-1

19. [slides 5-6] Given the following parameters, calculate the VaR of a position at 99% confidence interval with a holding period of 1 day:

    Number of call options = 50,000, current stock price = \$50,
    delta of the call option = 55%, volatility of stock σ =25%

    Assume that there are 250 trading days in one year and 99% probability = 2.33 standard deviations.

20. [c.f. slide 24] Using a parametric simulation approach, the risk of a complex portfolio shows the following projected profit/loss distribution for a holding period of one day:

Probability	1%	2%	10%	87%
Profit / Loss	Loss \$15 million	Loss \$3 million	Loss \$500,000	Profit > \$300,000

    a. What is the 95% VaR for the position?
    b. What is the 95% CVaR for the position?

Topic 3-2

21. [slides 5,8] The following table provides the historical return of a portfolio of US stocks (Portfolio A), together with data of the market return (S&P 500) and the risk free asset R_f:

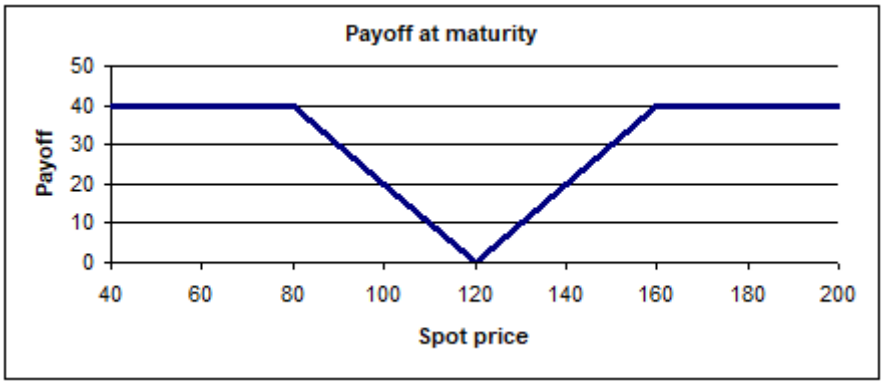
	Return R	Standard deviation of return	Beta
Portfolio A	5.5%	15%	0.60
S&P 500 index	7%	11%	1.00
Risk free asset R_f	2%	n/a	n/a

a. Calculate the Sharpe and Treynor measures for both Portfolio A and the S&P 500 index.
b. Form the Capital Allocation Line (CAL) with Portfolio A. Find the expected return of a portfolio P* with the same standard deviation of return as the S&P 500 index. Hence deduce the M^2 measure of Portfolio A.

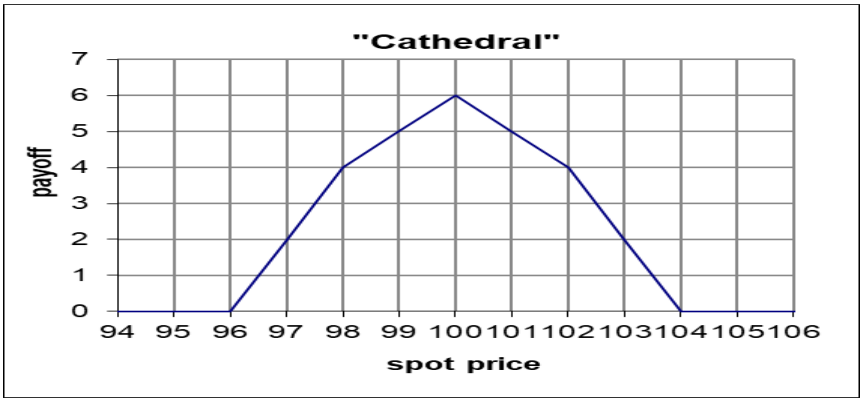
22. [slides 32, 44, 45, 46] Assume the following market parameters:
Current index level = 23754 points, interest rate = 4.13%p.a.
a. Maturity = 24 days (or 24/365 year), present value of expected dividends = 73 points. If the futures contract is trading at 23793, describe an arbitrage strategy using this futures contract and the stock basket. Calculate the theoretical profit (before transaction costs) and briefly discuss how the profit can be realized.
b. Maturity = 54 days (or 54/365 year), present value of expected dividends = 122 points. If the futures contract is trading at 23751, describe a trading strategy using this contract and the contract in (a) above. Calculate the potential profit (before transaction costs) and briefly discuss how the profit can be realized.

Examples of trading strategies

1. [c.f. Topic 1-1, slides 45ff]
   a. Using European calls and/or European puts only, suggest one way to obtain the following payoff pattern (payoff profile at maturity).



   b. Using European calls only, suggest one way to obtain the following payoff pattern (payoff profile at maturity).



2. Arbitrage strategy examples:
   a. Interest rate parity: see Topic 1-1, Q3 above.
   b. Put-call parity: see Topic 1-1, Q4 above.
   c. Index arbitrage: see Topic 3-2, Q22 above.

3. [Topic 1-4, slide 30; also see slide 33] Hedging strategy design
   An investment bank sold 10000 call options of stock A on Day 1. The following table shows the stock prices and the model-calculated option characteristics at the market close on the first three days (price, delta and vega are the parameters for 1 option).

	Stock price	Option price	Option delta	Option vega
Day 1	$75.3	$5.272	0.5662	0.147
Day 2	$72.4	$3.742	0.4671	0.142
Day 3	$76.5	$5.913	0.6049	0.145

   If the investment bank wants to keep the portfolio delta neutral at the end of each day, describe the trading actions that are needed at the end of each of the first three days. (No fractional shares can be traded; the minimum number of shares traded is 1 share; no need to discuss the treatment of the account balance in this question).

Indicative answers

1. (a) 8.633%; b) 8.281%

2. Theoretical futures price = US$4879.90.

3. Theoretical forward FX rate = 1.0855, which is different from the market forward FX rate (=1.0927). The arbitrage strategy consists of 4 trades at time t=0:
(1) borrow USD 1.0782 for 6 months
(2) spot FX transaction, sell USD 1.0782 to obtain EUR 1. [settled at time t=0]
(3) deposit EUR 1 for 6 months
(4) forward FX, to sell EUR 1.01875 and obtain (1.01875 x1.0927)=USD 1.11319
After 6 months, (3) would get EUR (1+0.0375 x 0.5) = 1.01875, which can be used to settle (4) and obtain USD 1.11319. After repaying the loan in (1), (1.0782 x (1+0.0513x0.5))=1.10586, the net profit is USD 1.11319 – 1.10586 = USD 0.00733.

4. c+Kexp(-rT) = $52.76, p+S0 = $52.26.
Therefore the strategy is sell call, buy put, buy stock, and the net payment upfront is $46.93. This amount is borrowed until maturity (3 months), and the amount that has to be returned is 46.93 x exp(0.0478 x 0.25) = $47.4942. Since $48 is received when the option is exercised either buy the buyer or the seller (----- you should supply your explanation here, following the reasoning in the lecture pdf-----), the net profit is $48-47.4942 = $0.5058 at maturity. An alternative strategy is, instead of borrowing $46.93, borrow an amount that would become $48 at maturity, so the net borrowing at time t=0 is 48 x exp(-0.0478 x 0.25) = $47.4298. The net upfront profit is thus $47.4298 – 46.93 = $0.4998. (---- you still need to explain what would happen at maturity ----).

5. Pricing by expectation: 0.75 x 100 + 0.25 x 70 = $92.5
Pricing by no arbitrage: 88 x exp(0.482 x 0.5) = $90.147

6. Fair price = price of zero coupon bond – option price = 99.4613%, to be paid by the bank to the investor. When a commission is charged by the bank, the investor will receive less money. The new note price is thus 99.4613%-0.30% = 99.1613%.

7. Fair price of the CGN = PV of G + participation x option premium = 97.242%. Since the selling price is 100%, the commission charged by the bank is 100%-97.242% = 2.758%.

8. The bank would want to replace this option by an option with a cheaper theoretical value. There are two possible ways to achieve this: either by reducing the payout (say from 2% to 1.8%), or by increasing the difficulty for the investor to receive the payout (say by changing the strike from 107% to 109%).

9. a = –0.3883 , b = –0.5915 , c = –0.3985.

10. a. estimated P&L = -2 x (-4300) + 2 x (-4000) = $600.
b. given the big change in the spot price, it is likely that the delta would change, and using delta to estimate the P&L is no longer accurate. An adjustment based on gamma should be applied.

11. a. portfolio s.d. = 24.16%
b. portfolio s.d. = 9.2%
c. $w_1 = 0.621, w_2 = 0.379$

12. Use portfolio s.d. to calculate the weight of A = 0.63636. Knowing weight of A (and weight of B = 1 – weight of A), $r_P = 9.273\%$.

13. a. Certainty equivalent of scheme A = $67600.
b. Expected utility of scheme B = $246.31, therefore A should be chosen.

14. Differentiate the expression, so that $36 - 4x = 0$, or $x = 9$. The function is only applicable for $0 \leq x \leq 9$.

15. a. $r_M = 10.26\%$
b. $\bar{r_2} = 11.95\%$

16. a. CML fair return = 12.583%. A lies below the CML.
b. SML fair return = 9.75%. A lies above the SML.

17. Required return from SML is 17.51%. Project expected return (=18%) is higher, therefore the project should be accepted.

18. According to CAPM, fair price of stock $\bar{Q}$ is $111.524. Since the market price (=$114.8) is trading above the fair price, it is not a good investment opportunity.

19. Equivalent notional amount = $2500000, adjusted by delta to become $1375000. VaR is $50655.74.

20. a. 95% VaR = $500000.
b. 95% CvaR = (1% x 15 + 2% x 3 + 2% x 0.5)/ 5% = $4400000.

21. a. Sharpe ratio: Portfolio A = 0.233, S&P 500 index = 0.4545
Treynor ratio: Portfolio A = 0.0583, S&P 500 index = 0.0500
b. Capital Allocation Line: 0.02+(0.055-0.02)/0.15*sigma = 0.02+0.2333*sigma
Return of P* = 4.567%, M^2 = –2.433%

22. a. Fair value of the futures contract = 23745.31. Since the market futures price (=23793) is above the theoretical futures price, regular arbitrage should be carried out. The strategy is: sell futures, borrow money to buy stocks. Theoretical profit is 23793-23745.31 = 47.69 points. [---Profit realization is discussed in the lecture pdf---]
b. Fair value of this futures contract = 23776.49. Since the market futures price (=23751) is below the theoretical futures price, one trading strategy is to buy this

futures contract and sell the contract from (a) above. The potential profit is the sum between the differences in the fair values and the futures contracts (47.69+23776.49-23751 = 73.08 points). However, there is no guarantee that a profit can be made [---Profit realization is discussed in the lecture pdf---]

Trading strategies

1.a. Buy a call spread (buy call at strike 120, sell call at strike 160) + buy a put spread (buy put at strike 120 + sell put at strike 80).

1.b. Long 2 calls (strike 96), long 2 calls (strike 104), short 1 call (strike 98), short 2 calls (strike 100), short 1 call (strike 102)

3. Day 1: buy 5662 shares at price $75.3.
Day 2: sell (5662–4671) = 991 shares at price $72.4.
Day 3: buy (6049–4671) = 1378 shares at price $76.5.
[option vega is not needed].